08/10/24

PROBABILITY

S / Ω = Sample Space

Emperical Definition:

Experimental definition

$\Omega = \{HH, HT, TH, TT\}$

$$P(A) = \frac{\#\text{ of favourable outcomes}}{\text{total }\#\text{ of outcomes}}$$

Trivial :

Non-trivial:

* [When sample space is uncountable]

classical definition does not work

=> axiomatic definition:

* Field :

i) if $A, B \in f$, then $A \cup B \in f$ and $A \cap B \in f$

$f = \{\phi, \Omega, \{A\}, \{A^c\}\}$

ii) if $A \in f$, then $A^c \in f$

$f = \{\phi, \Omega, \{A \cup B\}, \{A^c\}, \{A\}, \}$

A collection f of subsets of Ω is called a sigma field, if it satisfies following conditions:

i) $\phi \in f$

ii) if $A_1, A_2, A_3, \ldots\ldots \in f$ then

$$\bigcup_{i=1}^{\infty} A_i \in f$$

iii) if $A \in f$ then $A^c \in f$

* $\{\Omega, f, P\} \equiv$ Probability space

$$P : f \to [0,1]$$

i) $P(\phi) = 0$, $P(\Omega) = 1$

ii) If $A_1, A_2, A_3, \ldots\ldots$ is a collection of <u>disjoint</u> events of f such that $A_i \cap A_j = \phi$, if $i \neq j$, then $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$

i) $P(A^c) = 1 - P(A)$

ii) If $B \supseteq A$ then,

$$P(B) = P(A) + P(B \setminus A), \quad P(B) \geq P(A)$$

iii) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

iv) If $A_1, A_2, \ldots\ldots A_n$ be events,

then, $P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i) - \sum_{i<j} P(A_i \cap A_j)$

$$+ \sum P(A_i \cap A_j \cap A_k) - \ldots\ldots$$

$$\ldots + (-1)^{n+1} P(A_1 \cap A_2 \cap \ldots \cap A_n)$$

* $A \cup A^c = \Omega$

$$P(A) + P(A^c) = P(\Omega) = 1$$

$$P(A^c) = 1 - P(A^c)$$

*

$$A \cup (B \setminus A) = B$$

$$(A \subseteq B)$$

$$\Downarrow$$

$$P(A) + P(B \setminus A) = P(B)$$

Since, A & $B \setminus A$ are disjoint sets

$$P(B) - P(A) = P(B \setminus A) \geq 0$$

$$P(B) \geq P(A)$$

$\cancel{(A \cup B) + (A \cap B) = A + B}$

$\{\}$
$(\,)$

* $A \cup B = (A \setminus B) \cup (A \cap B) \cup (B \setminus A)$

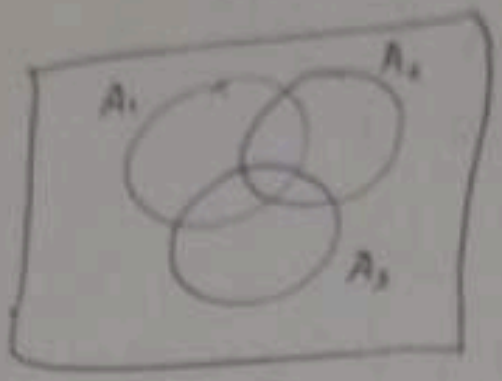
$$P(A \cup B) = P(A \setminus B) + P(A \cap B) + P(B \setminus A)$$

$$= P(A \setminus \overline{A \cap B}) + P(A \cap B) + P(B \setminus \overline{A \cap B})$$

$$= P(A) + P(B) - P(A \cap B) \Leftarrow = P(A) - P(A \cap B) + P(A \cap B) + P(B) - P(A \cap B)$$

* $P(A_1 \cup A_2 \cup A_3) \Rightarrow$

$$\cancel{A_1 \cup A_2 \cup A_3 = (A_1 \cap A_2 \cap A_3) \cup ((A_1 - (A_1 \cap A_2)) - (A_1 \cap A_3)) + (A_1}$$



$$P(A_1 \cup \overline{A_2 \cup A_3}) = P(A_1) + P(A_2 \cup A_3) - P(A_1 \cap (A_2 \cup A_3))$$

$$= P(A_1) + P(A_2) + P(A_3) - P(A_2 \cap A_3) - P((A_1 \cap A_2) \cup (A_1 \cap A_3))$$

$$= P(A_1) + P(A_2) + P(A_3) - P(A_2 \cap A_3) - P(A_1 \cap A_2) - P(A_1 \cap A_3) + P(A_1 \cap A_2 \cap A_3)$$

Let $A_1, A_2, \ldots$ be increasing sequence of event and let $A_1 \subseteq A_2 \subseteq A_3 \subseteq \ldots$ and write A

$\Rightarrow$ for the limit $A = \bigcup_{i=1}^{\infty} A_i = \lim_{i \to \infty} A_i$, then

$\Rightarrow$ show that, $P(A) = \lim_{i \to \infty} P(A_i)$.

$\Rightarrow$ Similarly, if $B_1, B_2, \ldots$ be decreasing sequence of events so that $B_1 \supseteq B_2 \supseteq B_3 \ldots$ and write the $B = \bigcap_{i=1}^{\infty} B_i = \lim_{i \to \infty} B_i$

$\rightarrow$ then show that $P(B) = \lim_{i \to \infty} P(B_i)$.

$\Rightarrow$ for $k = k'$,

Let, $P(A_1 \cup A_2 \cup \ldots \cup A_{k'})$

$$= \sum_{i=1}^{k'} P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \sum_{i<j<k} P(A_i \cap A_j \cap A_k)$$

$$+ \ldots + (-1)^{k'+1} P(A_1 \cap A_2 \cap \ldots \cap A_{k'})$$

$\Rightarrow$ for $k = k'+1$,

To prove, $P(A_1 \cup A_2 \cup \ldots \cup A_{k'} \cup A_{k'+1})$

$$= \sum_{i=1}^{k'+1} P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \sum_{i<j<k} P(A_i \cap A_j \cap A_k)$$

$$+ \ldots + (-1)^{k+2} P(A_1 \cap A_2 \cap \ldots \cap A_{k'} \cap A_{k'+1})$$

* Now, $P(A_1 \cup A_2 \cup A_3 \cdots \cup A_{k'} \cup A_{k'+1})$

$$= P(A_1 \cup A_2 \cup A_3 \cdots \cup A_{k'}) + P(A_{k'+1})$$

$$- P((A_1 \cup A_2 \cup A_3 \cdots \cup A_{k'}) \cap A_{k'+1})$$

$$= \sum_{i=1}^{k'} P(A_i) - \sum_{i<j<k'} P(A_i \cap A_j) + \sum_{i<j<k<k'} P(A_i \cap A_j \cap$$

$$\cdots + (-1)^{k'+1} P(A_1 \cap A_2 \cdots \cap A_{k'}) \quad [B_i = A_i \cap A$$

$$+ P(A_{k'+1}) - P(B_1 \cup B_2 \cup B_3 \cdots \cup B$$

$$\left\{ + P(B_1 \cup B_2 \cup \cdots \cup B_{k'}) \right.$$

$$= \sum_{i=1}^{k'} P(B_i) - \sum_{i<j} P(B_i \cap B_j)$$

$$+ \sum_{i<j<k} P(B_i \cap B_j \cap B_k)$$

$$\cdots + (-1)^{k'+1} P(B_1 \cap B_2 \cap B_3 \cdots \cap B_{k'}$$

$$\Rightarrow P(A_1 \cup A_2 \cup A_3 \cdots + A_{k'} \cup A_{k'+1})$$

$$= \sum_{i=1}^{k'+1} P(A_i) - \sum_{i<j<k'} P(A_i \cap A_j) - \sum_{i=1}^{k'} P(A_i \cap$$

$$+ \sum_{i<j<k<k'} P(A_i \cap A_j \cap A_k) + \sum_{i<j} P(A_i \cap A_j \cap A_{k'+1})$$

$$\left\{ + (-1)^{k'+1} P(A_1 \cap A_2 \cdots\cdots \cap A_{k'}) \right.$$

$$-(-1)^{k'} \sum_{i=1}^{k'} P\Big(\bigcap_{\substack{j=1 \\ j \neq i}}^{k'} A_j\Big) \cap A_{k'+1})$$

$$-(-1)^{k'+1} P(\cancel{B_1 \cap B}\ A_1 \cap A_2 \cap A_3 \cdots \cap A_{k'} \cap A_{k'+1})$$

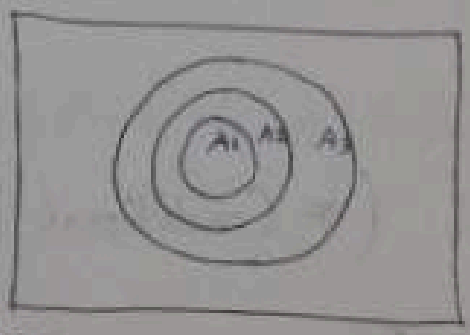
$$= \sum_{i=1}^{k'+1} P(A_i) - \sum_{i<j<k'+1} P(A_i \cap A_j) + \sum_{i<j<k<k'+1} P(A_i \cap A_j \cap A_k)$$

$$\cdots + (-1)^{k'+1} \sum_{i=1}^{k'+1} P\Big(\bigcap_{\substack{j=1 \\ j \neq i}}^{k'+1} A_j\Big)$$

$$+ (-1)^{k'+2} P(A_1 \cap A_2 \cap A_3 \cdots\cdots \cap A_{k'} \cap A_{k'+1})$$

$$= \underline{\overline{RHS}} \qquad \underline{\overline{H.P.}} \quad //$$

$$A = A_1 \cup (A_2 - A_1) \cup (A_3 - A_2) \cup \cdots\cdots$$



$$P(A) = P(A_1) + P(A_2 - A_1) + P(A_3 - A_2) + \cdots\cdots$$

$$= \cancel{P(A_1)} + \cancel{P(A_2)} - \cancel{P(A_1)} + \cancel{P(A_3)} - \cancel{P(A_2)} + \cdots\cdots$$

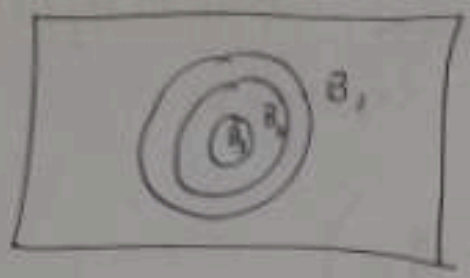
$$= \lim_{i \to \infty} P(A_i)$$

$B_1^c \subseteq B_2^c \subseteq \text{------}$

$$B^c = \bigcup_{i=1}^{\infty} B_i^c \quad \cancel{\lim_{i\to\infty} P(B_i^c)}$$

$$\boxed{P(B) = \lim_{i\to\infty} P(B_i)}$$

$$P(B^c) = \lim_{i\to\infty} P(B_i^c)$$

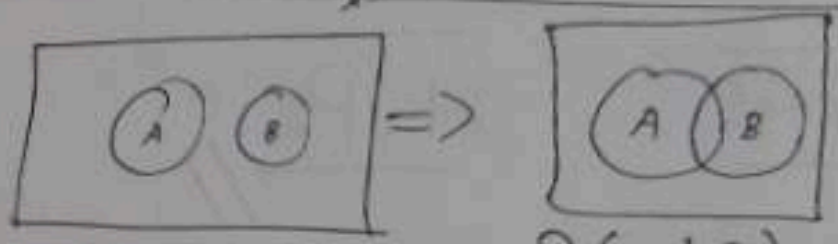


$$1 - P(B) = \lim_{i\to\infty} (1 - P(B_i))$$

$$\Rightarrow$$

$$P(B) = \lim_{i\to\infty} P(B_i)$$

\#



Chance of A if B occurs, $P(A|B) = \dfrac{P(A\cap B)}{P(B)}$

(A given B)

(* Conditional Probability,)

Q.11 For any events A, B such that $0 < P(B) < 1$

$$P(A) = P(A|B)\,P(B) + P(A|B^c)\,P(B^c)$$

$$\boxed{A = (A \cap B) \cup (A \cap B^c)}$$

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

$$P(A) = \frac{P(A \cap B)}{P(B)} P(B) + \frac{P(A \cap B^c)}{P(B^c)} P(B^c)$$

$$\Rightarrow P(A) = P(A|B)\,P(B) + P(A|B^c)\,P(B^c)$$

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* Let $B_1, B_2, \ldots, B_n$ be a partition of Ω such that $P(B_i) > 0 \ \forall i$.

mutually exclusive & jointly exhaustive

$$\bigcup_{i=1}^{n} B_i = \Omega$$

$$B_i \cap B_j = \phi \quad (\forall i, j)$$

Then, $P(A) = \sum_{i=1}^{n} P(A|B_i)\,P(B_i)$

$$A = (A \cap B_1) \cup (A \cap B_2) \cup (A \cap B_3) \ldots \cup (A \cap B_n)$$

$$\Rightarrow P(A) = P(A \cap B_1) + P(A \cap B_2) + \ldots + P(A \cap B_n)$$

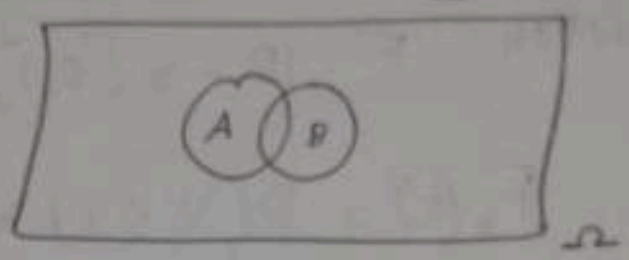
$$\Rightarrow P(A) = \cancel{P(A \cap B_i)} \sum_{i=1}^{n} P(A|B_i)\,P(B_i)$$

Bayes' Formula:

If $A_1, A_2, \ldots A_n$, is a partition of Ω, such that $\boxed{P(A_i) \neq 0 \ \forall i}$ A_i having positive prob., then

$$P(A_j|B) = \frac{P(B|A_j)\,P(A_j)}{\sum_{j=1}^{n} P(B|A_j)\,P(A_j)}$$

⇒ Independent events:



$$\frac{P(A \cap B)}{P(B)} = P(A \mid B) = P(A)$$

$$\frac{P(A \cap B)}{P(A)} = P(B \mid A) = P(B)$$

$$P(A \cap B) = P(A)P(B)$$

⇒ Random Variable: for countable with some prob
for uncountable density

A random variable is a function $X : \Omega \longrightarrow R$
with the property $\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}$ (sigma f))
$\forall x \in \mathbb{R}$.

$X \equiv$ no. of heads when ~~2 to~~ 2 times tossed

(probability mass function)

$$X = \begin{cases} 0 \longrightarrow \{T.T\} & \text{with prob. } \frac{1}{4} \\ 1 \longrightarrow \{HT, TH\} & \text{with prob. } \frac{1}{2} \\ 2 \longrightarrow \{HH\} & \text{with prob. } \frac{1}{4} \end{cases}$$

$\Rightarrow$ <u>Distribution Function:</u>

The distribution function of a random variable X is a function $F : \mathbb{R} \longrightarrow [0,1]$

$$F_X(x) = P(X \leq x) \ \forall x \in \mathbb{R}$$

$$\Rightarrow F_X(x) = \begin{cases} 0 \longrightarrow \text{when } x < 0 \\ 1/4 \longrightarrow \text{when } 0 \leq x < 1 \\ 3/4 \longrightarrow \text{when } 1 \leq x < 2 \\ 1 \longrightarrow \text{when } x \geq 2 \end{cases}$$

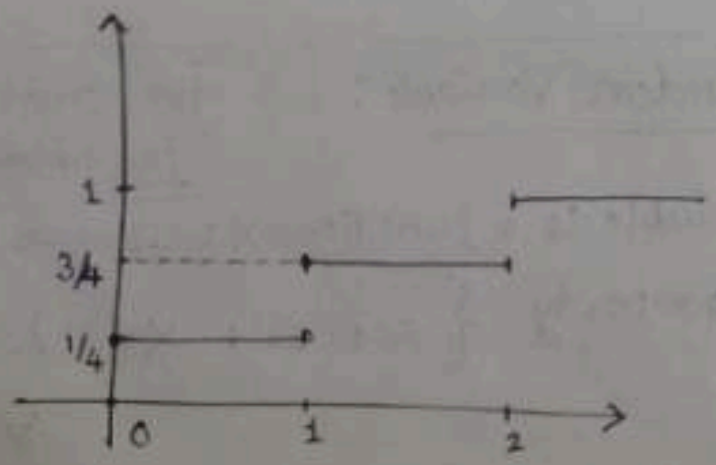
$\frac{1}{4}$ $\frac{1}{2}$ $\frac{1}{4}$

$F_X(1.5) = P(X \leq$

$= P(X=0) + P(X=$

$= \frac{1}{4} + \frac{1}{2}$

$= \frac{3}{4}$ //



i) $\lim_{x \to -\infty} F(x) = 0$, $\lim_{x \to \infty} F(x) = 1$

ii) If $x < y$ ~~where~~ then

$$F(x) \leq F(y)$$

iii) F is right continuous function

$*B_n = \{\omega \in \Omega : X(\omega) \leq -n\}$, $n = 1, 2, \ldots$

$$B_1 \supseteq B_2 \supseteq B_3 \supseteq \ldots$$

$$\lim_{n \to \infty} P(B_n) = P(\phi) = 0$$

$$\lim_{n \to -\infty} F(x) = 0 \;//$$

$*A_n$ 是 $= \{\omega \in \Omega : X(\omega) \leq n\}$, $n = 1, 2, \ldots$

$$A_1 \subseteq A_2 \subseteq A_3 \subseteq \ldots$$

$$\lim_{n \to \infty} P(A_n) = P(\Omega) = 1$$

$$\lim_{n \to \infty} F(x) = 1$$

$A_x = \{\omega \in \Omega : X(\omega) \leq x\}$

$\cancel{B_n = \{\omega \in \Omega : X(\omega) \leq y\}}$ $(y > x)$

$A_y = \{\omega \in \Omega : X(\omega) \leq y\}$

$A_{xy} = \{\omega \in \Omega : x < X(\omega) \leq y\}$

$$A_y = A_x \cup A_{xy}$$

$$P(A_y) = P(A_x) + P(A_{xy})$$

$$F(y) = F(x) + P(x < X \leq y)$$

$$* \; F(y) - F(x) = P(x < X \leq y)$$

$$\geq 0$$

$$\Rightarrow F(y) \geq F(x)$$

$A_n = \{ \omega \in \Omega : X(\omega) \leq x + \frac{1}{n} \}$,

$n = 1, 2, \ldots$

$B_n = \{ \omega \in \Omega : X(\omega) \leq x - \frac{1}{n} \}$, $n = 1, 2, \ldots$

$$* \; \lim_{n \to \infty} P(A_n) = P(X \leq x) = F(x)$$

$$\lim_{h \to 0^+} F(x+h) = F(x)$$

$$* \; \lim_{n \to \infty} P(B_n) = P(X < x)$$

$$\lim_{h \to 0^+} F(x-h) = P(X \leq x) - P(X = x)$$

$$= F(x) - P(X = x)$$

$F(x) = P(X \le x) = 1 - P(X > x)$

i) $P(X > x) = 1 - F(x)$

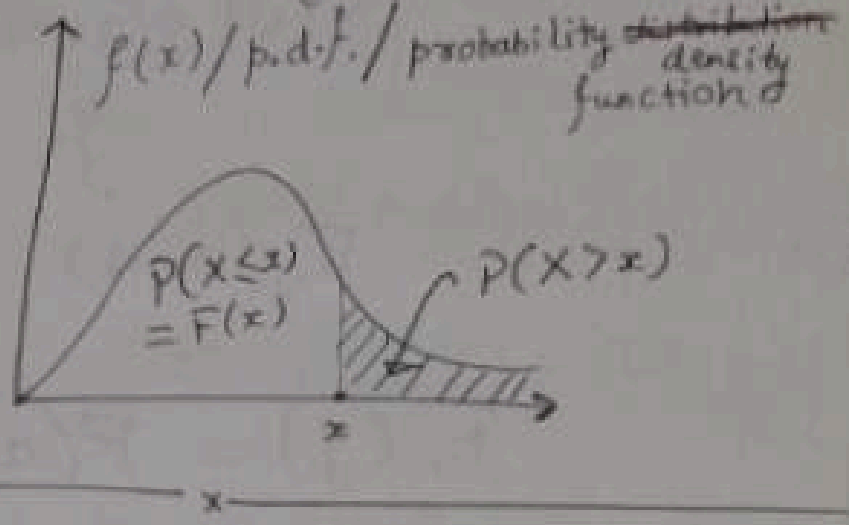
ii) $P(x < X \le y) = F(y) - F(x)$

iii) $P(X = x) = F(x) - \lim_{y \uparrow x} F(y)$

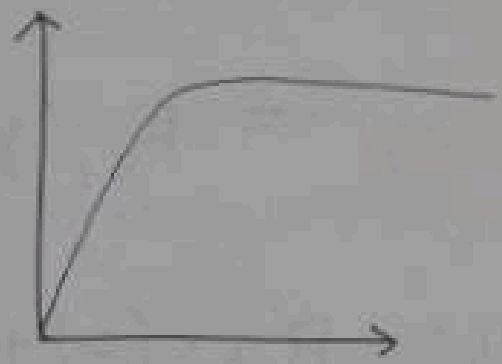
(y tends to x from below)

c.d.f / $F(x)$

cumulative distribution function



for continuous random variable, $P(X = x) = 0$.



Q.// A random variable X has distribution fun. $F_X(x)$. What is the distribution function of $Y = aX + b$, where $a, b \in \mathbb{R}$?

$F_Y(y)$

$F(aX(\omega) + b \le y)$

$X(\omega) \le x$

$aX(\omega) + b \le y$

$X(\omega) \le \frac{y-b}{a}$

$F_Y(y) = P(ax(\omega)+b \le y)$

$= P\left(x(\omega) \le \left(\frac{y-b}{a}\right)\right)$

$\overline{a>0}$

$= F_x\left(\frac{y-b}{a}\right)$

$\overline{a=0}$

$F_Y(y) = \begin{cases} 0, & \text{if } y<b \\ 1, & \text{if } y \ge b \end{cases}$

$\boxed{Y=b}$

$\overline{\text{if } a<0,}$

$F_Y(y) = P(Y \le y)$

$= P(ax+b \le y)$

$= P(-ax \ge b-y)$

$= 1 - P\left(x < \left(\frac{b-y}{-a}\right)\right)$

$= 1 - P\left(x \le \left(\frac{y-b}{a}\right)\right) + P\left(x = \frac{y-b}{a}\right)$

$= 1 - F_x\left(\frac{y-b}{a}\right) + P\left(x = \frac{y-b}{a}\right)$

Q11 i) S.T if F & G are distribution funⁿ,

then $\lambda F + (1-\lambda)G$ is a dist. funⁿ.

$0 \le \lambda \le 1$

ii) Is F G a distr. funⁿ.

=> To prove: non-decreasing prop. only.

$\lambda F + (1-\lambda)G = \lambda(F-G) + G$

$\overline{F-G>0}$

$0 \le \lambda(F-G) \le (F-G)$

$\left.\begin{array}{l} \lambda F + (1-\lambda)G \\ 0 \le \lambda \le 1 \\ 0 \le 1-\lambda \le 1 \end{array}\right\} \Rightarrow$ if F & G are inc.

$(\alpha F + \beta G)$, $\alpha \ge 0$, $\beta \ge 0$

will also be inc.

Consider, $x < y$

$$\lambda F(x) \leq \lambda F(y)$$

$$(1-\lambda) G(x) \leq (1-\lambda) G(y)$$

$$\lambda (F(x)) + (1-\lambda) G(x) \leq \lambda F(y) + (1-\lambda) G(y)$$

Consider, $x < y$

$$F(x) \leq F(y)$$

$$G(x) \leq G(y)$$

$$F(x) G(x) \leq F(y) G(y)$$

~~only when~~ ~~not valid when~~ ~~$F(x) < 0$ & $G(x)$~~

* Discrete Random Variable

The random variable X is called discrete if it takes values in some countable subset $\{x_1, x_2, \dots\}$ of R

The discrete random variable has probability mass function given by

$$f : \mathbb{R} \to [0, 1]$$

$$* \; f(x) = P(X = x), \; \forall x \in \mathbb{R}$$

The random variable 'X' is called continuous if it's distribution function can be expressed as $* \; F(x) = \int_{-\infty}^{x} f(x) \, dx$

for some integrable function $f: \mathbb{R} \to [0, \infty)$

Called the <u>probability density function</u>.

~~Joint~~ $\sum_{i=-\infty}^{\infty} f(x_i) = 1$

★ <u>Joint Distribution Function</u>:

The joint distribution function of a random vector,

$$\bar{X}(X_1, X_2, \ldots, X_n)$$

$$(\Omega, \mathcal{F}, P)$$

$$F_{\bar{X}}: \mathbb{R}^n \to [0,1]$$

$$F_{\bar{X}}(\bar{x}) = P(\bar{X} \leq \bar{x})$$

where $\bar{x} = (x_1, \ldots, x_n)$

$$= P(X_1 \leq x_1, X_2 \leq x_2, \ldots, X_n \leq x_n)$$

↳ The joint distribution fun^n $F_{X,Y}$ of the random vector (X, Y) have the following properties,

i) $\lim_{x,y \to -\infty} F_{X,Y}(x,y) = 0$, $\lim_{x,y \to \infty} F_{X,Y}(x,y) = 1$

ii) If $(x_1, y_1) \leq (x_2, y_2)$ then $F_{X,Y}(x_1, y_1) \leq F_{X,Y}(x_2, y_2)$

iii) $F_{X,Y}$ is continuous from above, i.e., $F_{X,Y}(x+u, y+v) \to F_{X,Y}(x,y)$ as $u, v \to 0^+$

* Joint Probability Mass Function :

The random variable X & Y on the probablity space $(\Omega, \mathcal{F}, P)$ are called jointly discrete if vector X, Y takes values in some countable subsets of R.

The jointly discrete variable X, Y have joint probability mass function given by,

$$f : \mathbb{R}^2 \to [0, 1]$$

$$f(x, y) = P(X = x \;\&\; Y = y)$$

$$= P(X = x, Y = y)$$

* Jointly continuous random variable :

The random variable X & Y on the probability space $[\Omega, \mathcal{F}, P]$ are called jointly continuous if their joint distribution function can be expressed as,

$$F_{X,Y}(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(u, v)\, dv\, du \quad \forall\, x, y \in \mathbb{R}$$

for some integrable function $f(x, y)$,

$f : \mathbb{R}^2 \to [0, \infty)$

$\Rightarrow$ joint probability density function

Independant Bernoulli Trials:

mean = value of central tendancy

dispersion =

variance =

$$X = \begin{cases} 1, & \text{with probability } p \\ 0, & \text{with probability } q = 1-p \end{cases}$$

$x_1, x_2, \ldots, x_n$

$$\text{mean/average} = \frac{\sum_{i=1}^{n} x_i}{n} \simeq \sum_{i=1}^{n} x_i f(x_i)$$

$$m_k = E[X^k] = \sum_x x^k f(x)$$

Moment of order k,

$$E[X] = \sum_x x f(x)$$

$k = 1, 2, 3, \ldots$

$$\sigma_k = E[(X - m_1)^k] = \sum_x (x - m_1)^k f(x)$$

Central moment of order k

mean ($k=1$) deviation		standard deviation $\sqrt{\sigma^2}$
	$k=2$	variance σ^2

$\sigma_2 = m_2 - m_1^2$

2nd moment

$$\sigma^2 = \sigma_2 = \text{Variance of } X = \text{Var}(X)$$

$$\sqrt{\sigma^2} = \sigma$$

$$E[x] = 1 \cdot p + 0 \cdot (1-p)$$
$$= p$$

$$\sigma_2 = \sum (x - m_1)^2 f(x)$$

Single trial

$$= \sum_x x^2 f(x) - 2m_1 \sum_x x f(x) + m_1^2 \sum_x f(x)$$

$$\boxed{\begin{array}{l} X = \begin{cases} 1 & , p \\ 0 & , 1-p \end{cases} \\ m_1 = E[x] = p \\ \boxed{m_2 = p} \\ \sigma_2 = m_2 - m_1^2 \\ = pq \end{array}}$$

$$= m_2 - 2m_1 m_1 + m_1^2$$

$$= m_2 - m_1^2$$

$$m_2 = E[x^2] = 1^2 \cdot p + 0^2 \cdot (1-p)$$
$$= p$$

* n Bernoulli Trials :

$x \equiv$ no. of heads,

$$f(x) = P(X = x) = ?$$

$$f(x) = {}^nC_x \, p^x (1-p)^{n-x} = {}^nC_k \, p^k (1-p)^{n-k},$$

$$k = 0, 1, 2, \ldots, n$$

$$E[x] = ? = \sum_x x^k f(x)$$

$$Var(x) = ? \qquad = \sum_{x=0}^{n} x \, {}^nC_x \, p^x (1-p)^{n-x}$$

$$\sigma_2 = \sigma^2 = E[(X - m)^2] = \sum_x (x - m_1)^2$$

$$E[x] = \sum_{k=1}^{n} k \frac{\lfloor n}{\lfloor k \, \lfloor n-k} p^k (1-p)^{n-k}$$

$$= np \sum_{k=1}^{n} \frac{\lfloor n-1}{\lfloor k-1 \, \lfloor n-1-(k-1)} p^{k-1} (1-p)^{n-k}$$

$$= np \, (p+1-p)^{n-1} = np$$

$$\sigma_2 = \sum_k (k - np)^2 \, {}^nC_k \, p^k (1-p)^{n-k}$$

$$= \sum_k k^2 \, {}^nC_k \, p^k (1-p)^{n-k}$$

$$+ -2np \sum_k k \, {}^nC_k \, p^k (1-p)^{n-k}$$

$$+ n^2p^2 \sum_k {}^nC_k \, p^k (1-p)^{n-k}$$

$$k^2 \frac{n(n-1)}{k(k-1)} \, {}^{n-2}C_{k-2}$$

$$nk({}^{n-1}C_{k-1})$$

$$n(n-1)$$

$$n((n-1)p^2 C_{k-2} + n-1 \lfloor_{k-1} =$$

$$-2np \, np + n^2p^2$$

$$= -n^2p^2$$

$$\Rightarrow \boxed{m_2 = E[x^2] = E[x(x-1)] + E[x] = \gamma_2 + m_1}$$

$$X \sim \text{binomial}(n, p)$$

$$\gamma_k = E[x(x-1)(x-2) \cdots (x-r+1)]$$

$$m_2 = E[x^2] = E[x(x-1)] + E[x]$$

$$= \gamma_2 + m_1$$

$$\gamma_2 = \sum_{k=0}^{n} k(k-1)\,{}^nC_k\, p^k (1-p)^{n-k}$$

$$= n(n-1)p^2 \sum_{k=2}^{n} \frac{(n-2)!\; p^{k-2}(1-p)^{n-k}}{(k-2)!\,(n-2-(k-2))!}$$

$$\Rightarrow \gamma_2 = n(n-1)p^2 (p+1-p)^{n-2} = n(n-1)$$

$$\Rightarrow m_2 = \cancel{n(m} \; n(n-1)p^2 + np$$

$m_2 = \gamma_2 + m_1$

$$= n^2p^2 - np^2 + np$$

$$\Rightarrow Var(X) = m_2 - m_1^2 = \cancel{n^2p^2} - np^2 + np - \cancel{n^2p^2}$$

$$= np(1-p)$$

$$= npq$$

$$\Rightarrow Sd(X) = \sqrt{npq}$$

$$\boxed{\begin{array}{l} \Rightarrow {}^nC_k\, p^k (1-p)^{n-k} \\ \Rightarrow np \\ \Rightarrow npq \\ \Rightarrow \gamma_2 = n(n-1)p^2 \end{array}}$$

Q.11 $X \overset{D}{\sim}$ binomial (n, p)

$$X = Y_1 + Y_2 + \cdots + Y_n$$

$$Y_i \overset{D}{\sim} \text{Bernoulli}(p) \Rightarrow, i = 1, 2, \cdots, n$$

$$E[ax + by] = ? \qquad \underline{a, b \in \mathbb{R}}$$

(*) Expectation operator have following properties.

i) $E[aX + bY] = ?$ $\qquad a, b \in \mathbb{R}$

ii) $E(\text{constant}) = \text{constant}$

iii) if $X \geq 0$, the $E[X] \geq 0$

iv) $\text{var}(aX + bY) = ?$

What will happen if X & Y are indepent?

$\Rightarrow X, Y$

$$\text{cov}(X, Y) = E[(X - m_x)(Y - m_y)]$$

$$E(X+Y) = \sum_{\omega \in \Omega} (X+Y)(\omega) P(\omega) = \sum_{\omega \in \Omega} X(\omega) P(\omega) + \sum_{\omega \in \Omega} Y(\omega) P(\omega)$$

$$= E(X) + E(Y)$$

(*) $\boxed{E[ax+by] = aE[x] + bE[y]}$

$$\left\{ E[ax^2+by^2+cz^2] = aE[x^2] + bE[y^2] + cE[z^2] \right\}$$

$a, b, c \in \mathbb{R}$

$$\Rightarrow E[aX+bY] = \sum_x \sum_y (ax+by) f(x,y)$$

$$= \sum_x ax \sum_y f(x,y) + \sum_y by \sum_x f(x,y)$$

$$= a \sum_x x f_X(x) + b \sum_y y f_Y(y)$$

$$= aE[X] + bE[Y]$$

(*) $f(x,y)$

$g(X,Y)$

$$E[g(X,Y)] = \sum_x \sum_y g(x,y) f(x,y)$$

$\sum_y f(x,y) = \sum_y P(X=x, Y=y)$

$$= P(X=x)$$

$$= f_X(x) \Longrightarrow \text{Marginal Mass Function}$$

$$\sigma_2 = E[(X-m_1)^2]$$

$$P(X=x) = P\left(\bigcup_y \{X=x, Y=y\}\right)$$

$$= \sum_y P(X=x, Y=y)$$

$$= \sum_y f_{X,Y}(x,y)$$

$$\circledast \ Var(aX+bY) = E\left[(Z-E[Z])^2\right]$$

$$= E\left[(aX+bY-aE[X]-bE[Y])^2\right]$$

$$= E\left[(a[X-E[X]] + b[Y-E[Y]])^2\right]$$

$$= E\left[a^2(X-E[X])^2 + b^2(Y-E[Y])^2 + 2ab(X-E[X])(Y-E[Y])\right]$$

$$= a^2 E\left[(X-E[X])^2\right] + b^2 E\left[(Y-E[Y])^2\right] + c^2 E[(Z-$$

$$= a^2 Var(X) + b^2 Var(Y) + 2ab\, Cov(X,Y)$$

$$\circledast \ Var(aX+bY+cZ) = a^2 Var(X) + b^2 Var(Y) + c^2 Var(Z) + 2ab\, Cov(X,Y) + 2bc\, Cov(Y,Z) + 2ac\, Cov(X,Z)$$

$\Rightarrow$ X & Y are two independant events,

then $f_{X,Y}(x,y) = f_X(x) f_Y(y) \quad \forall x,y$

$$\begin{cases} P(X=x, Y=y) = P(X=x)\, P(Y=y) \\ P(A \cap B) = P(A)\, P(B) \end{cases}$$

$\hookrightarrow Cov(X,Y) = E\left[(X-E[X])(Y-E[Y])\right]$

$$= E\left[XY - YE[X] - XE[Y] - E[X]E[$$

$$= E[XY] - E[X]E[Y] - E[X]E[Y] + E[X]E[Y]$$

$$= E[XY] - E[X]E[Y]$$

$$\Rightarrow \quad E[XY] = \sum_x \sum_y xy f_{X,Y}(x,y)$$

$$= \sum_x \sum_y xy\, f_X(x) f_Y(y)$$

$$= \sum_x x\, f_X(x) \sum_y y f_Y(y)$$

$$= E[X]E[Y]$$

$\Rightarrow$

$$X = \begin{cases} -1, & \text{with prob. } 1/3 \\ 0, & \text{with prob. } 1/3 \\ 1, & \text{with prob. } 1/3 \end{cases}, \quad Y = \begin{cases} 1, & \text{with prob. } \frac{2}{3} \\ 0, & \text{" " } \frac{1}{3} \end{cases}$$

$\boxed{Y = X^2}$

$$\left.\begin{aligned} P(X=-1, Y=1) &= 1/3 \\ P(X=-1) &= 1/3 \\ P(Y=1) &= \frac{2}{3} \end{aligned}\right\}$$

$$E[XY] = E[X^3]$$

$$= -1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3}$$

$$= 0$$

$$E[X] = 0, \quad E[Y] = \frac{2}{3}$$

$\#$ $Y \overset{D}{\sim} \text{binomial}(n, p)$

$$Y = X_1 + X_2 + \cdots + X_n$$

$$X_i \overset{D}{\sim} \text{Bernoulli}(p)$$

$$\Rightarrow E[Y] = np, \quad Var(Y) = npq, \quad \overline{q = 1-p}$$

$X \sim \text{Poisson}(\lambda)$

$$\boxed{\sum_{i=1}^{\infty} k f(k)}$$

$$E[x] = \sum_{i=1}^{\infty} i \frac{\lambda^i e^{-\lambda}}{\lfloor i}$$

$$\frac{\lambda^k e^{-\lambda}}{\lfloor k}$$

$$= e^{-\lambda} \sum_{i=1}^{\infty} \frac{\lambda^i}{\lfloor i-1} = \lambda e^{-\lambda} \sum_{i=1}^{\infty} \frac{\lambda^{i-1}}{\lfloor i-1}$$

$$= \lambda e^{-\lambda} e^{\lambda} = \lambda_{//}$$

$$Var(x) = m_2 - m_1^2$$

$$Var(x) = m_2 - m_1^2$$

$$m_2 = m_1 + \gamma_2$$

$$\boxed{\begin{array}{l} \frac{\lambda^k e^{-\lambda}}{\lambda^2 \lfloor k} \\ np = \lambda \\ = np = \lambda \end{array}}$$

$$m_2 = E[x^2] = \sum_{i=1}^{\infty} i^2 \frac{\lambda^i e^{-\lambda}}{\lfloor i}$$

$$= \sum_{i=1}^{\infty} i \frac{\lambda^i e^{-\lambda}}{\lfloor i-1} = e^{-\lambda} \sum_{i=1}^{\infty} \frac{d(\lambda^i)}{d\lambda}$$

$$\frac{i \lambda^{i-1}}{\lfloor i-1}$$

$$= e^{-\lambda} \frac{d e^{\lambda}}{d\lambda} \times \lambda = \lambda_{//}$$

$$e^{\frac{d\lambda}{d\lambda}}$$

$$e^{\lambda} = \sum_{i=0} \frac{i \lambda^{i-1}}{(i-1)!} = \sum_{i=1} \frac{i \lambda^{i-1}}{(i)!} = \sum \frac{\lambda^{i-1}}{(i-1)!}$$

$$\cancel{Var(x) = \lambda + \lambda^2 - \lambda^2}$$

$$m_2 = E[x^2] = E[x(x-1)] + E[x]$$

$$\Rightarrow \gamma_2 = E[x(x-1)] = \sum_{i=2}^{\infty} i(i-1) \frac{e^{-\lambda} \lambda^i}{\lfloor i}$$

$$= e^{-\lambda} \lambda^2 \sum_{i=2}^{\infty} \frac{\lambda^{i-2}}{\lfloor i-2} = \lambda^2 e^{-\lambda} e^{\lambda} = \lambda^2_{//}$$

(waiting time to get the 1st success)

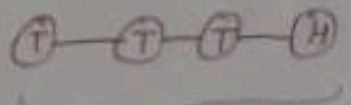
time till the success

(no. of trails till the success)

↳ Geometric Random Variable:

$$X \sim \text{geometric}(p)$$

$$P(X=k) = P(\{H_1=T\} \cap \{H_2=T\} \cap \{H_3=T\} \cap \cdots \cap \{H_k=$$



$X=4$

$$= (1-p)^{k-1} p,$$

$k = 1, 2, 3, \ldots$

$E[X] = m_1 = \sum_{i=1}^{\infty} i f(i)$

$$= \sum_{i=1}^{\infty} i (1-p)^{i-1} p$$

$$= p \sum_{i=1}^{\infty} i (1-p)^{i-1}$$

$$= p(-1) \sum_{i=1}^{\infty} \frac{d(1-p)^i}{dp} = \cancel{-p}$$

$$(-p) \frac{d}{dp} \sum_{i=1}^{\infty} (1-$$

$$= -p \frac{d}{dp} \frac{(1-p)}{1-(1-p)}$$

$$= -p \frac{d}{dp} \left(\frac{1-p}{p} \right)$$

$$= -p \frac{d}{dp}\left(\frac{1}{p}-1\right) = -p \times \left(\frac{-1}{p^2}\right) = \frac{1}{p} //$$

$\underline{\#}$

$$E[x^2] = E[x(x-1)] + E[x]$$

$$\boxed{\begin{array}{l} (1-p)^n p \\ \text{mean} = \frac{1}{p} \\ \text{Var} = \frac{q}{p^2} \\ \gamma_2 = \frac{3q}{p^2} \end{array}}$$

$$m_2 = \sum_{i=1}^{\infty} i^2 f(i) \Rightarrow \quad m_2 = \gamma_2 + m$$

$$\gamma_2 = \sum_{i=1}^{\infty} i(i-1) f(i)$$

$$= \sum_{i=1}^{\infty} i(i-1)(1-p)^{i-1} p$$

$$= p(1-p) \sum_{i=1}^{\infty} \frac{d^2}{dp^2}\left((1-p)^{i-2}\right)$$

$$= p(1-p) \frac{d^2}{dp^2}\left(\frac{1-p}{p}\right) = p(1-p) \frac{2}{p^3}$$

$$\Rightarrow \gamma_2 = \frac{2p(1-p)}{p^2}$$

$$\text{Var}(x) = \frac{2}{p^2} - \frac{2}{p} + \frac{1}{p} - \frac{1}{p^2}$$

$$= \frac{1}{p^2} - \frac{1}{p} = \frac{1}{p}\left(\frac{1}{p}-1\right) = \frac{1}{p^2} - \frac{1}{p} //$$

$$\Rightarrow \frac{1}{p}\left(\frac{1}{p}-1\right)$$

$$= \frac{q}{p^2} //$$

$E[x] = \sum_{k=1}^{\infty} k(1-p)^{k-1} p$

$$= p \sum_{k=1}^{\infty} k(1-p)^{k-1} \Rightarrow p \frac{1}{(1-1+p)^2} = \frac{1}{p}$$

For $|x| < 1$, $\frac{1}{1-x} = 1 + x + x^2 + \cdots$

$\Rightarrow \frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \cdots$

$$\Rightarrow \frac{2}{(1-x)^3} = 2 + 6x + 12x^2 + \cdots$$

$E[x^2] = E[x(x-1)] + E[x]$

$$\Rightarrow \gamma_2 = E[x(x-1)] = \sum_{k=1}^{\infty} k(k-1)(1-p)^{k-1} p$$

$$= p(1-p) \sum_{k=1}^{\infty} k(k-1)(1-p)^{k-2}$$

$$= p(1-p) \frac{2}{(1-(1-p))^3} = \frac{2(1-p)}{p^2}$$

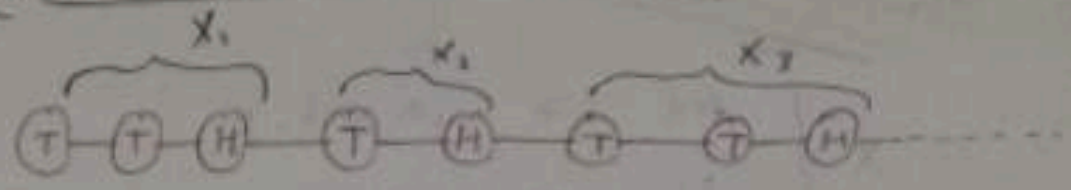
$$\Rightarrow m_2 = \gamma_2 + m_1 = \frac{2(1-p)}{p^2} + \frac{1}{p} = \frac{2}{p^2} - \frac{1}{p}$$

$$\Rightarrow \sigma_2 = Var(x) = m_2 - m_1^2 = \frac{2}{p^2} - \frac{1}{p} - \frac{1}{p^2} = \frac{1}{p^2} - \frac{1}{p}$$

$$\Rightarrow \boxed{Var(x) = \frac{1-p}{p^2}}$$

(probability till the nth success)

$\Rightarrow$



$$\Rightarrow X = X_1 + X_2 + \cdots + X_r$$

$$X_i \overset{D}{\sim} \text{geometric}(p)$$

k^{th} toss
r^{th} success

$$P(X=k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k = r, r+1, \ldots$$

$$\Rightarrow m_1 = E[X] = \sum_{i=1}^{\infty} {}^{i-1}C_{r-1}\, p^r (1-p)^{i-r}$$

$\binom{k-1}{r-1} p^r (1-p)^{k-r}$
mean $= \frac{r}{p}$
Var $= \frac{r(1-p)}{p^2}$
$\gamma_2 = \frac{r[1-2p]}{p^2}$

$$= p^r \sum_{i=1}^{\infty} p^{r-1}\, {}^{i-1}C_{r-1} (1-p)^{i-r}$$

$$\cancel{= p^r (1-p) = p \,(}$$

$$= p \lim_{i \to \infty} (p + 1 - p)^{i-1}$$

$$= p$$

$$\overset{\#}{X} = X_1 + X_2 + \cdots + X_r$$

$$E[X] = E[X_1] + E[X_2] + \cdots + E[X_r] = \frac{r}{p}$$

$$\cancel{P(X=x)}$$

$$Var[X] = Var[X_1] + Var[X_2] + \cdots + Var[X_r]$$

$$= \frac{r(1-p)}{p^2}$$

Trinomial Distribution:

#

$$X \begin{cases} \text{red} \rightarrow \text{with prob. } p \\ \text{blue} \rightarrow \text{with prob. } q \\ \text{green} \rightarrow \text{with prob. } (1-p-q) \end{cases}$$

$$0 < p+q < 1$$

$X = \#$ of red outcomes in n trials

$Y = \#$ of blue outcomes in n trials

$$\Rightarrow \boxed{P(X=i, Y=j) = \frac{\lfloor n}{\lfloor i \, \lfloor j \, \lfloor n-i-j} p^i q^j (1-p-q)^{n-i-j}}$$

$$\Rightarrow P(X=i) \Rightarrow P\left(\bigcup_{j=0}^{n-i} \{X=i, Y=j\}\right)$$

$$\cancel{F(x=k) = \sum_{n=-\infty}^{k} P(X=x, Y=j)}$$

$$\cancel{= \frac{q^j}{(1-p-k)^j}} = \sum_{j=0}^{n-i} P\{X=i, Y=j\}$$

$$= \sum_{j=0}^{n-i} \frac{\lfloor n}{\lfloor i \, \lfloor j \, \lfloor n-i-j} p^i q^j (1-p-q)^{n-i-j}$$

$$= \frac{\lfloor n}{\lfloor i \, \lfloor n-i} \sum_{j=0}^{n-i} \frac{\lfloor n-i}{\lfloor j \, \lfloor n-i-j} q^j (1-p-q)^{n-i-j}$$

$$= \frac{\lfloor n}{\lfloor i \, \lfloor n-i} \; p^i (q+1-p-q)^{n-i} = \frac{\lfloor n}{\lfloor i \, \lfloor n-i} \times p^i (1-p)^{n-i}$$

$$\Rightarrow P(X=i) = {}^nC_i \; p^i (1-p)^{n-i}$$

$X \overset{D}{\sim} \text{binomial}(n, p)$

$Y \overset{D}{\sim} \text{binomial}(n, q)$

$\hookrightarrow$ $P(X=i, Y=j) = P(X=i)\, P(Y=j)$

$$\Downarrow$$

$\hookrightarrow$ $f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y) \;\; \forall i, j$

$$\Downarrow$$

$$F_{X,Y}(x,y) = F_X(x)\, F_Y(y) \;\; \forall x, y$$

$$\Rightarrow P(X \le x, Y \le y) = P(X \le x) \cdot P(Y \le y)$$

$$\Downarrow$$

$$\Rightarrow F_{X,Y}(x,y) = F_X(x)\, F_Y(y)$$

$$\Downarrow$$

$$\Rightarrow f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y)$$

$\hookrightarrow$ $f(x,y) = \dfrac{\alpha^x \beta^y}{\lfloor x \, \lfloor y} e^{-\alpha-\beta}$, $x = 0, 1, 2, \ldots$, $y = 0, 1, 2, \ldots$, $\alpha, \beta > 0$

$$f_X(x) = \cancel{\sum_{y=0}^{n-x}} \frac{\alpha^x}{\lfloor x} e^{-\alpha}, \quad f_Y(y) = \frac{\alpha^y}{\lfloor y} e^{-\beta}$$

(*) $P(X=k) = {}^{n}C_{k}\, p^{k} (1-p)^{n-k}$

In binomial dis
X, Y are depend
when n is fixe

$P(Y=j) = {}^{n}C_{j} (1-p)^{j} p^{n-j}$

$P(X=k, Y=j) = 0 \quad , \; j \neq n-k$ //

(*) $N \overset{D}{\sim} \text{Poisson}(\lambda)$

when $n \overset{D}{\sim}$ Poisson(λ)
the independent

$P(B/A) = \dfrac{P(B \cap A)}{P(A)}$,

$\hookrightarrow P(X=i, Y=j) = P(X=i) \cdot P(Y=j) \quad \forall i, j$

$N \overset{D}{\sim} \text{Poisson}(\lambda)$

$* \; P(X=i) = P\left(\bigcup_{k=0}^{\infty} \{X=i, N=i+k\}\right)$

$= \sum_{k=0}^{\infty} P(X=i, N=i+k) = \sum_{k=0}^{\infty} P(X=i / N=i+k) P(N$

$\Downarrow$ $P((X=i) \cap (N=i+k))$

$= \sum_{k=0}^{\infty} {}^{i+k}C_{i}\, p^{i} (1-p)^{k} \dfrac{e^{-\lambda} \lambda^{i+k}}{\lfloor i+k}$

$= e^{-\lambda} \sum_{k=0}^{\infty} \dfrac{\lfloor i+k}{\lfloor i \, \lfloor k} p^{i} (1-p)^{k} \dfrac{\lambda^{i+k}}{\lfloor i+k} = \dfrac{e^{-\lambda p} (\lambda p)^{i}}{\lfloor i}$

$X \overset{D}{\sim} \text{Poisson}(\lambda p)$

$Y \overset{D}{\sim} \text{Poisson}(\lambda(1-p))$

$\Rightarrow P(X=i, Y=j) = P(X=i, Y=j, N=i+j)$

$= P(X=i, Y=j \mid N=i+j) \cdot P(N=i+j) = {}^{i+j}C_i \, p^i (1-p)^j \dfrac{e^{-\lambda} \lambda^{i+j}}{\lfloor i+j}$

$= \dfrac{e^{-\lambda} (\lambda p)^i (\lambda(1-p))^j}{\lfloor i \, \lfloor j}$

$\Rightarrow$ Eq ① = Eq ②, $\therefore$ X, Y both are independant.

⊛ Symmetric Random Variable:

We say a random variable X, is symmetric about a ~~random~~ point ~~variable~~ α,

$1 - P(X \geq \alpha - x)$

$P(X \geq \alpha + x) = P(X \leq \alpha - x) \quad \forall x \in \mathbb{R}$

$= 1 - P(X < \alpha + x) = 1 - P(X \leq \alpha + x) + P(X = \alpha + x)$

$F(\alpha - x) = 1 - F(\alpha + x) + P\{X = \alpha + x\} \; \forall x \in \mathbb{R}$

$\alpha =$ center of symmetry

If $\alpha = 0 \Rightarrow F(-x) = 1 - F(x) + P\{X = x\}$

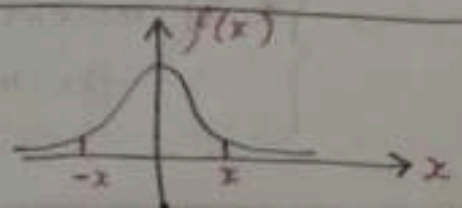
$\Rightarrow$ If X is continuous r.v.

$f(\alpha - x) = f(\alpha + x) \; \forall x \in \mathbb{R}$

$\therefore$ If $\alpha = 0$, then $f(-x) = f(x) \; \forall x \in \mathbb{R}$

⊛ pdf of cauchy random variable:

$f(x) = \dfrac{1}{\pi} + \dfrac{1}{1+x^2}, \quad -\alpha < x < \alpha,$



$$F(x) = \int_{-\infty}^{x} \frac{1}{\pi} \frac{1}{1+x^2}\, dx = \frac{1}{\pi}\left[\tan^{-1}(x)\right]_{-\infty}^{x}$$

$$F(x) = \frac{1}{\pi}\tan^{-1}(x) + \frac{1}{2} \quad //$$

$$E[X] = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{x}{1+x^2}\, dx$$

$E[X^k]$ does not exist $k \geq 1$

$\hookrightarrow$ $$\lim_{b\to\infty,\, a\to\infty} \frac{1}{\pi}\int_{-a}^{b} \frac{x}{1+x^2}\, dx \text{ d.n.e.}$$

$$\lim_{a\to\infty} \frac{1}{\pi}\int_{-a}^{a} \frac{x}{1+x^2}\, dx = 0$$

$$\frac{1}{\pi}\int_{-a}^{0} \frac{x}{1+x^2}\, dx = \lim_{a\to\infty} \frac{1}{2\pi}\left[\ln(1+x^2)\right]$$

$$= \lim_{a\to\infty} \frac{1}{2\pi}\ln(1+a^2) \text{ dne}$$

$$E[|x|^s] = \int_{|x|^s \leq 1} |x|^s f(x)\, dx + \int_{|x|^s > 1} |x|^s f(x)\, dx$$

$$\leq P\{|x|^s \leq 1\} + E[|x|^t] < \infty$$

{* If moment of order k exists the moment $\leq t$ also exist}

$$E[(x-c)^k] \Rightarrow \sigma_k = E[(x-m_1)^k]$$

$$= E\left[x^k - \binom{k}{1} x^{k-1} m_1 + \cdots\cdots\right]$$

$$= m_k - \binom{k}{1} m_1 m_{k-1} + \binom{k}{2} m_1^2 m_{k-2} - \cdots\cdots$$

$$\cdots\cdots + (-1)^k m_1^k$$

$$\Rightarrow \quad m_k = E[x^k] = E[(x - m_1 + m_1)^k]$$

$$= \sigma_k + \binom{k}{1} m_1 \sigma_{k-1} + \binom{k}{2} m_1^2 \sigma_{k-2} + \cdots\cdots + m_1^k \sigma_0$$

Quantile

$P\{x \le x\} \ge p$, $P\{x \ge x\} \ge 1-p$, $0 < p < 1$

* A quantile of order p is equal to $100p^{th}$ percentile.

* A no. of x satisfying, $P(X \le x) \ge p$, $P(X \ge x) \ge 1-p$ is called quantile of order p ($100 \cdot p^{th}$ percentile)

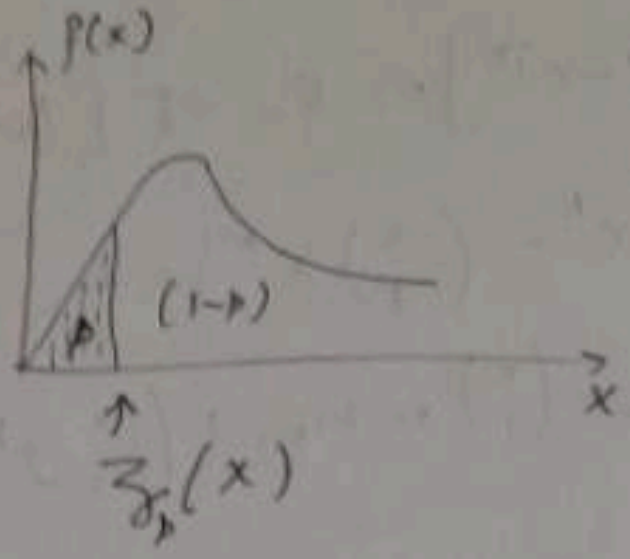
$$p \le F(x) \le p + P\{X = x\}$$

$$F(x) \ge p, \quad \left\{ \begin{array}{l} 1 - P(X < x) \ge 1-p \\ 1 - F(x) + P\{X = x\} \ge 1-p \\ F(x) \le p + P\{X = x\} \end{array} \right\}$$

* x is quantile of order p for some r.v. X having D.F. F.

If x is cont. random variable $F(x) = p$,

Give $\xi_p(X)$ (quantile of order p) = all solutions of x



Median:

Let X be a RV with DF, F(x), a number x such that

$$\frac{1}{2} \leq F(x) \leq \frac{1}{2} + P\{X = x\}$$

(or), $P(X \leq x) \geq \frac{1}{2}$ and $P\{X \geq x\} \geq \frac{1}{2}$

↳ is called median of F(x)

(can have multiple satisfying values)

$$P\{X = -2\} = P\{X = 0\} = \frac{1}{4}$$

$$P\{X = 1\} = \frac{1}{3}, \quad P\{X = 2\} = \frac{1}{6}$$

For Cauchy
r.v. $F(x) = 1/2$
$(x = 0) \Rightarrow$ median

$$F(x) = \begin{cases} 0 & x < -2 \\ 1/4 & x < 0 \\ 1/2 & x < 1 \\ 5/6 & x < 2 \\ 1 & x > 2 \end{cases}$$

$$\begin{cases} \sum_j \sum_k x^j y^k f(x,y) = m_{jk} = E[X^j Y^k] \\ \qquad\qquad\qquad\qquad = \iint x^j y^k f(x,y)\, dx\, dy \\ \sigma_{jk} = E[(X - E[X])^j (Y - E[Y])^k] \end{cases}$$

$m_{10} = E[X]$, $m_{01} = E[X]$

$m_{20} = E[X^2]$, $m_{02} = E[Y^2]$, $m_{11} = E[XY]$

$$\begin{cases} \sigma_{10} = \sigma_{01} = 0, \ \sigma_{20} = Var(X), \ \sigma_{02} = Var(Y) \\ \sigma_{11} = Cov(X,Y) \end{cases}$$

<u>Prove:</u> $Var(X) < E[(X-c)^2]$ for any $c \neq E[X]$

$\mu = E[X]$

$$Var(X) = E[(X-\mu)^2] = E[(X-c)+(c-\mu)]^2$$

$$= E[(X-c)^2] + E[(c-\mu)^2] + E[2(c-\mu)(X-c)]$$

$$= E[(X-c)^2] - (c-\mu)^2$$

$Var(X) - E[(X-c)^2] = -(c-\mu)^2 < 0 \Rightarrow \underline{Var(X) < E(}$

To find relationship b/w 2 R.V., when density is given:

Correlation coefficient $\underbrace{0 < \mu < 1}_{\text{perfectly linear}}$

$\Rightarrow Y \approx ax + b \Rightarrow$

$b = E[Y - ax]$
$= E[Y] - a$

$$E[(Y - ax - b)^2]$$

$$= E[(Y - ax - E[Y] + aE[X])^2]$$

$$= E[\{(y - E[Y]) - a(x - E[X])\}^2]$$

$$= \sigma_y^2 + a^2\sigma_x^2 - 2a\,Cov(X,Y)$$

$$\boxed{Var(X) = \sigma_x^2, \quad a = \frac{Cov(X,Y)}{\sigma_x^2}}$$

$$\min_{a,b} E[(Y - ax - b)^2] = \min_a \{\sigma_y^2 - 2a\,cov(X,Y) + a^2\sigma_x^2\}$$

$$= \frac{\sigma_x^2\sigma_Y^2 - \{cov(X,Y)\}^2}{\sigma_x^2}$$

$$= \sigma_Y^2\left\{1 - \left[\frac{Cov(X,Y)}{\sigma_x\sigma_Y}\right]^2\right\}$$

$$= \sigma_y^2 \{1 - \rho^2\}$$

$$\Rightarrow \rho = \frac{Cov(X,Y)}{\sigma_x \sigma_y}$$

$$1 - \rho^2 \geq 0 \Rightarrow \rho^2 \leq 1, \quad -1 \leq \rho \leq 1$$

Correlation Coefficient

$\rho \equiv$ measure of degree of linearity b/w X & Y

$\rho^2 \equiv$ Coefficient of determination

(*) Correlation Coefficient :

$$f(x,y) = \frac{\lfloor n}{\lfloor x \, \lfloor y \, \lfloor n-x-y} p^x q^y (1-p-q)^{n-x-y}$$

$x, y = 0, 1, 2, \ldots, n$

$x + y \leq n$

$p + q < 1$

$$\rho = \frac{Cov(X,Y)}{\sigma_x \sigma_y}$$

$$= \frac{E[XY] - E[X]E[Y]}{\sqrt{Var(X) \cdot Var(Y)}}$$

$E[XY] = E[X]E[Y]$

$\Uparrow$

independent

$p + q < 1$

$X \sim$ binomial (n, q)

$Y \sim$ binomial (n, p)

$$= \frac{\cancel{np\,nq}}{\cancel{npq}\ \cancel{npq}} \qquad = \frac{\cancel{np\,nq}}{\sqrt{\cancel{npq}\ \cancel{np}}}$$

$$= \frac{-nq\,np}{\sqrt{np(1-p)\ nq(1-q)}}$$

$$E[XY] = \sum_{x=0}^{n} \sum_{y=0}^{n-x} xy \frac{\lfloor n}{\lfloor x \lfloor y \lfloor n-x-y} p^x q^y (1-p-q)^{n-x-y}$$

$$= q \sum_{x=0}^{n} \frac{x \lfloor n}{\lfloor x \lfloor n-x} p^x \sum_{y=1}^{n-x} \frac{y \lfloor n-x}{\lfloor y \lfloor n-x-y} q^{y-1} (1-p-q)^{n-x-y}$$

$$= q \sum_{x=0}^{n} \frac{\lfloor n}{\lfloor x-1 \lfloor n-x-1} p^x \sum_{y=1}^{n-x} \frac{\lfloor n-x-1}{\lfloor y-1 \lfloor n-x-1-y-1} q^{y-1} (1-p-q)^{n-x-y}$$

$$= q \sum_{x=1}^{n} \frac{\lfloor n}{\lfloor x-1 \lfloor n-x-1} p^x (q+1-p-q)^{n-x-1}$$

$$= pq \sum_{x=1}^{n} \frac{\lfloor n}{\lfloor x-1 \lfloor n-x-1} p^{x-1} (1-p)^{n-x-1}$$

$$= n(n-1)pq \sum_{x=1}^{n} \frac{\lfloor n-q}{\lfloor x-1 \lfloor n-q-x-1} p^{x-1} (1-p)^{n-q-x}$$

$$= n(n-1)pq \quad //$$

$$\Rightarrow \frac{n(n-1)pq - npnq}{\sqrt{np(1-p)\,nq(1-q)}} = \frac{-npq}{\sqrt{n^2(1-p)(1-q)pq}} = -\sqrt{\frac{p}{(1-p)}}$$

Conditional Distribution ;

$$F_{Y|X}(y|x) = P(Y \le y \mid X = x) = \frac{P(Y \le y, X = x)}{P(X = x)}$$

'0' for continuous

$$f_{Y|X}(y|x) = P(Y = y \mid X = x)$$

$$= \frac{P(Y = y, X = x)}{P(X = x)} = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

$$f_{X|Y}(x|y) = P(X = x \mid Y = y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$E[X] \equiv$ constant

$E[Y|X] \equiv$ variable $\equiv$ Random variable

$$\psi(x) = E[Y|X] = \sum_y y f_{Y|X}(y|x)$$

RV with mass function $f(x)$

$$E[E[Y|X]] = ? = E[\psi(x)] = \sum_x \psi(x) f_X(x)$$

$$= \sum_x \sum_y y f_{Y|X}(y|x) f_X(x)$$

$$= \sum_x \sum_y y \frac{f_{X,Y}(x,y)}{f_X(x)} f_X(x) = \sum_y y \sum_x f_{X,Y}(x,y)$$

$$= \sum_y y f_Y(y) = E[Y] \,//$$

$$E[x^i y^j] = E_x[x^i \, E_{y|x}[Y^j]]$$

$$= \sum_x \sum_y x^i y^j f_{x,y}(x,y) = \sum_x x^i f_x(x) \sum_{(y|x)} y^j f_{y|x}(y)$$

$$f_{x|y}(x|y) = \frac{f_{x,y}(x,y)}{f_y(y)}$$

$$= \frac{\frac{\cancel{\lfloor n}}{\lfloor x \, \lfloor y \, \lfloor n-x-y} \cancel{p^x q^y} (1-p-q)^{n-x-y}}{\frac{\cancel{\lfloor n}}{\lfloor y \, \lfloor n-y} \cancel{q^y} (1-q)^{n-y}}$$

$$= \frac{\lfloor n-y}{\lfloor x \, \lfloor n-y-x} \left(\frac{p}{1-q}\right)^x \left(\frac{1-p-q}{1-q}\right)^{n-x-y}, \quad x = 0, 1, \ldots$$

$$E[X|Y] = \frac{(n-y)p}{1-q}, \quad E[E[X|Y]] = \frac{n}{}$$

$$= \frac{np(1-q)}{(1-q)} = np$$

$$\frac{-p}{1-q} n$$

$$x = 0, 1, \ldots$$

$\Rightarrow$ binomial $\left(n-y, \frac{p}{1-q}\right)$

$$\boxed{E[X|Y] = \frac{(n-y)p}{1-q}}$$

$$\cancel{E[X^2Y^2] = E_X[X^2 E}$$

$$E[X\, E_{Y|X}[Y]]$$

$$= \frac{nq\,E[X]}{1-p} - \frac{p\,E[X^2]}{1-p}$$

(*) $E[XY] = E_Y[\,Y E_{X|Y}[X]]$,

$$= \frac{npE[Y]}{1-q} - \frac{p\,E[Y^2]}{1-q} = E_Y\left[\frac{Y[n-Y]p}{1-q}\right]$$

$$= \frac{n^2pq}{1-q} - \frac{p[nq(1-q) + n^2q^2]}{1-q}$$

$$= \frac{nq\,np}{1-p} - \frac{q[np(1-p) + n^2p^2]}{1-p}$$

$$= \frac{n^2pq - npq + np^2q - n^2p^2q}{1-p}$$

$$= \frac{n^2pq}{1-q} - npq - \frac{n^2q^2p}{1-q} = \frac{n^2pq}{\cancel{(1-q)}}\cancel{(1-q)} - npq$$

$$= n(n-1)pq$$

$$n^2pq - npq = n(n-1)pq$$

---

(2) An urn contain 6 red, 8 green & 6 blue balls, pick 24 balls at random with replacement.

Let $R, G, B \rightarrow$ be the random variables counting the no. of red, green & blue balls resp.

(a) Find joint pmf of (R, G, B)

Sol^n:

$$\frac{24!}{r!\,g!\,(24-r-g)!} \times P(\text{red ball drawn})\; P(\text{green ball drawn})$$

$$f(x,y) = \frac{24!}{x!\,y!\,(24-x-y)!} \times \left(\frac{6}{24}\right)^x \left(\frac{8}{24}\right)^y \left(\frac{6}{24}\right)^{24-x-y} P(\text{blue ball drawn})$$

#

$\Rightarrow Var(Y) = E[Var(Y|X)] + Var(E[Y|X])$

$$\left\{ Var[Y|X] = E[Y^2|X] - \left\{E[Y|X]\right\}^2 \right\}$$

$\hookrightarrow E[E[Y^2|X]] = E[Y^2]$

① $\Rightarrow E[Var[Y|X]] = E[Y^2] - E[(E[Y|X])^2]$

② $\Rightarrow Var(E[Y|X]) = E[\{E[Y|X]\}^2] - \{E[Y]\}^2$

Adding ① + ②, we obtain,

$$E[Var[Y|X]] + Var[E[Y|X]]$$

$$= E[Y^2] - \{E[Y]\}^2$$

$$= Var(X)$$

$E[(Z - E[Z])^2]$

$X \overset{D}{\sim} Poisson(\lambda)$, $Y \overset{D}{\sim} Poisson(\mu)$ } independant

Find the dist. of the r.v. $X+Y$?

Also, find the dist. of $X \mid X+Y=n$?

$\hookrightarrow Z = X+Y$, $E[Z] = E[X] + E[Y]$

$$Var(Z) = Var(X) + Var(Y)$$

$$f_Z(z) = P(Z=z) = P\left(\bigcup_x \{X=x,\ Y=z-x\}\right)$$

$$= \sum_x P\{X=x, Y=z-x\} = \sum_x P\{X=x\}P(Y=z-x)$$

$$= \sum_{x=0}^{z} \frac{e^{-\lambda}\lambda^x}{\lfloor x} \frac{e^{-\mu}(\mu)^{z-x}}{\lfloor z-x} = \frac{e^{-\lambda-\mu}}{\lfloor z} \sum_{x=0}^{z} \frac{\lfloor z}{\lfloor x \lfloor z-x} \lambda^x \mu^{z-x}$$

$$f_Z(z) = \frac{e^{-(\lambda+\mu)}}{\lfloor z} \cdot (\lambda+\mu)^z$$

$$\boxed{Z \overset{D}{\sim} \text{Poisson}(\lambda+\mu)}$$

$$\Rightarrow P(X=x \mid X+Y=n) = \frac{P(X=x, X+Y=n)}{P(X+Y=n)}$$

$$= \frac{P(X=x)\,P(Y=n-x)}{P(X+Y=n)} = \frac{\dfrac{e^{-\lambda}\lambda^x}{\lfloor x} \dfrac{e^{-\mu}(\mu)^{n-x}}{\lfloor n-x}}{\dfrac{e^{-(\lambda+\mu)}(\lambda+\mu)^n}{\lfloor n}}$$

$$= \frac{\lfloor n}{\lfloor x \lfloor n-x} \cdot \frac{\lambda^x (\mu)^{n-x}}{(\lambda+\mu)^n}$$

$$= \frac{\lfloor n}{\lfloor x \lfloor n-x} \frac{(\lambda)^x}{(\lambda+\mu)^x} \frac{(\mu)^{n-x}}{(\lambda+\mu)^{n-x}} \Rightarrow X \mid X+Y=n \overset{D}{\sim}$$

$$\text{binomial}\left(n, \frac{\lambda}{\lambda+\mu}\right)$$

Q.11 The life time of a machine is a random variable T with pmf f.

Given that, the machine is working in after t days, what is the mean subsequent life time of the machine?

when, $f(x) = \begin{cases} \frac{1}{N+1}, & x = 0, 1, 2, \ldots, N \\ \frac{1}{2^x}, & x = 1, 2, \ldots \end{cases}$

$$E[x] = \sum x f(x) = \frac{N}{N+1}$$

$$E[X] = \sum x f(x) = \sum_{x=1}^{\infty} x \frac{1}{2^x} =$$

$$\left(\frac{1/2}{1 - 1/2}\right) \quad \left(\frac{x}{1-x}\right) =$$

Let T_1 = # of additional days the machine survives after t days

$$P(T_1 = r \mid T > t) = \frac{P(T_1 = r, T > t)}{P(T > t)}$$

$$= \frac{P(T = t+r, T > t)}{\left(\frac{N-t}{N+1}\right)} = \frac{P(T = t+r)}{\frac{N-t}{N+1}} = \frac{1}{N-t}$$

$$E[T_1 \mid T>t] = 1\cdot\frac{1}{N-t} + 2\cdot\frac{1}{N-t} + \cdots + (N-t)\frac{1}{(N-t)}$$

$$= \frac{1}{(N-t)}\,\frac{(N-t)(N-t+1)}{2} = \frac{(N-t+1)}{2} \quad //$$

$$P(T_1 = r \mid T>t) = \frac{P(T=t+r)}{P(T>t)} = \frac{P(T=t+r)}{1-P(T\le t)}$$

$$P(T\le t) = \sum_{r=1}^{t}\frac{1}{2^r} = \frac{\frac{1}{2}\left(1-\left(\frac{1}{2}\right)^t\right)}{\left(1-\frac{1}{2}\right)} = \left(1-\left(\frac{1}{2}\right)^t\right)$$

Dno. $\Rightarrow 1-\left(1-\left(\frac{1}{2}\right)^t\right) = \left(\frac{1}{2}\right)^t$ //

$$\Rightarrow P(T_1 = r \mid T>t) = \frac{1/2^{t+r}}{1/2^t} = \frac{1}{2^r} \quad //$$

$$E[T_1 \mid T>t] = \sum_{r=1}^{\infty} r\,\frac{1}{2^r} = 2$$

$$S = 1\cdot\frac{1}{2} + 2\cdot\frac{1}{2^2} + \cdots$$

$$\frac{1}{2}S = \frac{1}{2^2} + 2\cdot\frac{1}{2^3} + \cdots$$

$$\frac{1}{2}S = \frac{1}{2} + \left(\frac{1}{2^2} + \frac{1}{2^3} + \cdots\right) = \frac{1}{2} + \frac{1/2}{1-1/2} = 1$$

$$\boxed{S = 2} \quad //$$

Let x is a poisson random v. with parameter μ.

P.T. $P(x \leq n) = \frac{1}{n!} \int_{\mu}^{\infty} e^{-x} x^n dx$

$$\Rightarrow \text{Let } I_n = \int_{\mu}^{\infty} e^{-x} x^n dx = \left[\frac{x^n e^{-x}}{-1}\right]_{\mu}^{\infty} - \int_{\mu}^{\infty} \frac{n x^{n-1}}{-1}$$

$$= \mu^n e^{-\mu} + \int_{\mu}^{\infty} n x^{n-1} e^{-x} dx$$

$$= \mu^n e^{-\mu} + n \int_{\mu}^{\infty} x^{n-1} e^{-x} dx$$

$$= \mu^n e^{-\mu} + n \int_{\mu}^{\infty} x^{n-1} e^{-x} dx = \mu^n e^{-\mu} + I_{n-1} //$$

$$\Rightarrow I_n = e^{-\mu} \mu^n + n I_{n-1}$$

$$\Rightarrow \boxed{J_n = \frac{I_n}{n!}} \Rightarrow$$

$$\# \; J_n = \frac{e^{-\mu} \mu^n}{n!} + \frac{I_{n-1}}{(n-1)!} = P(X=n) + \cancel{P(X=n-1)}$$

$$= P(X=n) + P(X=n-1) + J_{n-2}$$

$$= P(X=n) + P(X=n-1) + P(X=n-2) + \text{------} + P(X=1) + P(X=$$

$$\therefore J_n = P(X \leq n)$$

$$\therefore P(X \leq n) = \frac{1}{n!} \int_{\mu}^{\infty} e^{-x} x^n dx //$$

$$J_0 = \frac{I_0}{0!} = I_0 = \int_{\mu}^{\infty} e^{-x} dx = e^{-\mu} //$$